Module 12: Darwin and Evolution — Questions

Quick Check

1. Based on his observations in the Galápagos Islands, Charles Darwin concluded that: A) Environment has no impact on physical traits. B) Organisms instinctively adapt to be perfect. C) Species can change over time as they adapt to different environments. D) All species were created at the exact same time and never change.

2. Which of the following is the best definition of "evolutionary fitness"? A) The physical strength and speed of an organism B) The ability of an individual to survive and produce viable, fertile offspring relative to others C) The overall health and immune system of an animal D) The dominance of an animal in its social group

3. In biology, natural selection acts on: A) Individuals, causing them to evolve during their lifetime B) Populations, causing genetic traits to shift over generations C) The environment, forcing it to change for the animals D) Acquired characteristics (like learning a behavior)

4. The forelimbs of a human, a cat, a whale, and a bat share the same internal bone structure, even though they are used for entirely different functions (grabbing, walking, swimming, flying). This is an example of: A) Analogous structures B) Vestigial structures C) Homologous structures D) Convergent evolution

5. True or False: Natural selection is a random process. - [ ] True - [ ] False

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Reflection

6. Why is the phrase "survival of the fittest" often misleading to the general public? Explain what fitness actually means in an evolutionary context.

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7. A population of beetles has both green and brown individuals. They live in a lush green forest, and birds frequently eat the brown beetles because they stand out. Over time, the

forest experiences a severe drought and the leaves turn brown. Explain step-by-step how the beetle population will evolve via natural selection in response to this environmental change.

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8. Briefly explain how molecular biology (DNA and proteins) provides evidence for evolution and common ancestry.

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